

Calculus

Quiz 1

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Time Limit: 20 mins

A final answer without showing your steps or reasoning will not receive any points. You don't have to answer the questions in order, but you must clearly label each question number and make sure the answer areas for different questions are distinct.

- ✓ 1. (20 pts) Retrieval practice: (a) Describe the definition of a **function** in your own words. (b) Give two examples where these mappings do not satisfy the definition, and explain why they do not.
- ✓ 2. (20 pts) Retrieval practice: Explain the definition and give an example of each type of functions: (i) odd function (ii) power function (iii) rational function (iv) decreasing function (v) transcendental function
- ✓ 3. (20 pts) Retrieval practice: (a) How to define the logarithmic function with a positive base $b \neq 1$? State its **domain** and **range**. (b) Prove the change of base formula: $\log_b(x) = \frac{\ln(x)}{\ln(b)}$.
- ✓ 4. (20 pts) Retrieval practice: (a) Describe a **process (method)** for the tangent function $\tan(x)$ having a well-defined **inverse function**. (b) Provide the graphs of $\tan(x)$ and $\tan^{-1}(x)$, and state their own domain-and-range.
- ✓ 5. (28 pts) Let $f(x) = 1/\sqrt{x+1}$ and $g(x) = x - 1$. Find the functions and their **domains** (i) $f \circ f$, (ii) $f \circ g$, (iii) $g \circ f$, and (iv) $g \circ g$.
6. (22 pts) Show your steps or reason to find the value of each:
✓ (i) $\arccsc(\sqrt{2}) = ?$ (ii) $\operatorname{arccot}(-\sqrt{3}) = ?$ (iii) $\cos(\sin^{-1}(\frac{\pi}{6})) = ?$

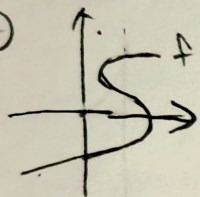
1. (a)

在 set A 中的 element, 能夠對應到唯一一個在 set B 中的 element. +8

(b)

① $x^2 + y^2 = 4$, $x=0$ 時, $y = \pm 2$, failed on vertical line test. +12

②



f 在同一 x 有 >1 個 y 值 ✓

2. (i) $f(x) = -f(x)$ Eg. $f(x) = x^3$ +20

(ii) $f(x) = x^n$, n 為 integer. Eg. $f(x) = x^2$ ✓

(iii) $f(x) = \frac{p(x)}{q(x)}$, p, q 為多項式 Eg. $f(x) = \frac{x^2 + x + 1}{x + 1}$ ✓

(iv) $f(x) \geq f(x+h)$, $h > 0$. Eg. $f(x) = -2x$ ✓

(v) 不能以代數運算達成的函式 ✓ Eg. $f(x) = \sin x$
多項式

3. (a) $\log_b x = y \Leftrightarrow b^y = x$ +10

(b) $\log_b(1) =$

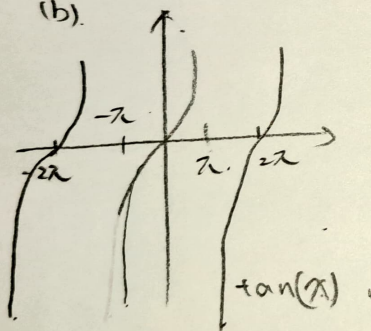
domain = $(0, \infty)$

range = $(-\infty, \infty)$

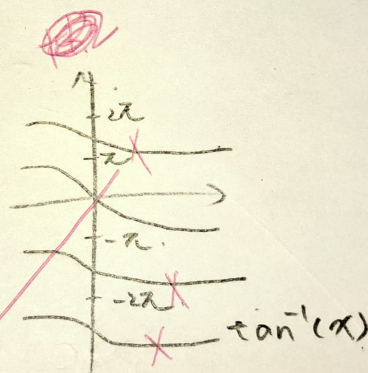
= $\mathbb{R}$

4. (a) $\tan^{-1}(\tan x) = x$ well-defined

(b).



+6



domain = $\{x \mid x \neq k\pi, x \in \mathbb{R}, k \in \mathbb{Z}\}$

range = $\{y = \tan(x) \mid y \in \mathbb{R}\}$

domain = $\{x \mid x \in \mathbb{R}\}$

Range = $\{y = \tan^{-1}(x) \mid y \neq k\pi, y \in \mathbb{R}, k \in \mathbb{Z}\}$

$\mathbb{R}?$

3.

4.

5.

(i)

$$(f \circ f)(x) = \frac{1}{\sqrt{\frac{1}{\sqrt{x+1}} + 1}}$$

$$x+1 > 0$$

$$x > -1$$

+28

$$(-1, \infty)$$

(ii)

$$(f \circ g)(x) = \frac{1}{\sqrt{x-1+1}} = \frac{1}{\sqrt{x}}, \quad (0, \infty)$$

(iii)

$$(g \circ f)(x) = \frac{1}{\sqrt{x+1}} - 1, \quad (-1, \infty)$$

$$x+1 > 0$$

$$x > -1$$

(iv)

$$(g \circ g)(x) = (x-1)-1 = x-2, \quad (-\infty, \infty)$$

IR

6.

(i)

+6

$$\cos k = \frac{1}{\sqrt{2}}$$

$$k = \frac{\pi}{4}$$

(ii)

$$\tan k = -\frac{1}{\sqrt{3}}$$

$$k = \frac{5\pi}{3}$$

5/6 TV

(iii)

$$\sin k = \frac{\pi}{6}$$

$$\frac{\pi}{3}$$

$$\sqrt{1 - \frac{\pi^2}{36}}$$

3. prove $\log_b(x) = \frac{\ln x}{\ln b}$

$$y = \log_b x \Rightarrow b^y = x$$

$$\log_a b^y = \log_a x$$

$$y \log_a b = \log_a x$$

$$y = \frac{\log_a x}{\log_a b} \Rightarrow \text{If } a = e, y = \frac{\ln x}{\ln b}$$

4. (a)

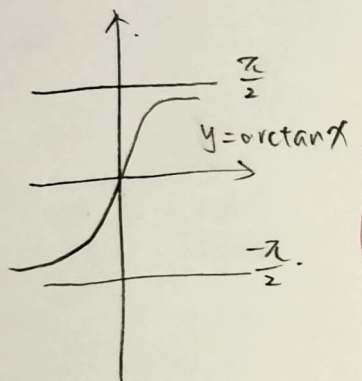
$$\langle \tan x \rangle$$

$$x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \text{ range} = \mathbb{R}$$

$$\langle \arctan x \rangle$$

$$x \in \mathbb{R}, \text{ range} = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

(b) $\langle \arctan x \rangle$



$$\text{domain} = \mathbb{R}$$

$$\text{range} = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

6 (i)

$$\arccot(-\sqrt{3}) =$$

$$\Rightarrow \frac{1}{\tan k} = -\sqrt{3}$$

$$\tan k = -\frac{1}{\sqrt{3}}, k = \frac{5}{6}\pi$$

$$-\frac{\pi}{6} + \pi = \frac{5}{6}\pi$$

(ii)

$$\cos(\sin^{-1}(\frac{\pi}{6}))$$

$$\sin k = \frac{\pi}{6}$$

$$\cos k = \sqrt{1 - \sin^2 k} = \sqrt{1 - \left(\frac{\pi}{6}\right)^2} = \sqrt{1 - \frac{\pi^2}{36}}$$